
Where the Hierarchy Breaks: Failure Motifs and Calibrated Uncertainty for HIP-NN Molecular Energies under Size Extrapolation

Anonymous Authors
Paper 1 follow-up study
prepared with the hippynn package (LANL)

Abstract

Machine-learning interatomic potentials (MLIPs) report sub-chemical-accuracy mean errors on molecular benchmarks, yet a small tail of large errors limits their use in molecular dynamics, transition-state searches, and screening. The Hierarchically Interacting Particle Neural Network (HIP-NN) decomposes molecular energy into per-atom contributions of increasing many-body order and exposes a built-in uncertainty signal, the non-hierarchicality R ; the original authors showed qualitatively that large R correlates with large error, but did not characterize the chemistry that drives the tail, whether R is quantitatively calibrated, or whether failures concentrate in a few interpretable motifs. We train HIP-NN with the original held-fixed architecture and training protocol on the ≤ 8 -heavy-atom molecules of QM9 and evaluate on the 9-heavy-atom remainder, a controlled size-extrapolation regime. On this out-of-distribution chemistry we (i) identify a compact library of *hierarchy-failure motifs* (HFM)—structural, electronic, and steric motifs, defined operationally from geometry alone, whose conditional failure rates exceed the background rate by factors of up to three—and (ii) build a calibrated uncertainty model (HCU) from hierarchy-failure signals (R , per-order energy magnitudes, and a cross-seed ensemble spread) whose predicted error is close to the identity line on held-out molecules (ECE ≈ 0.018 kcal/mol), with substantially better failure-detection discrimination than R alone. A Pareto analysis shows that the failure tail is compact rather than diluted: the top 4 motifs explain $\geq 80\%$ of identifiable failures. The mechanism splits into two regimes: the structural motifs fail *loudly*—their second-order ($n=2$) contributions fail to decay, so the model knows it is failing—while the delocalization motifs fail *quietly*, with near-normal hierarchies, exposing a blind spot of any hierarchy-based uncertainty estimate. A transfer test on 60,000 off-equilibrium conformations from ANI-1x shows which motifs are properties of the chemistry and which were artifacts of the minimized-geometry distribution. All analysis runs post hoc on a single trained model in seconds, adding negligible cost to a HIP-NN deployment.

1 Introduction

Machine-learning interatomic potentials map an atomic configuration $\mathcal{C} = \{(Z_i, \mathbf{r}_i)\}$ to a molecular energy $\hat{E}(\mathcal{C})$, and modern architectures reach mean absolute errors (MAE) well below the 1 kcal/mol threshold of “chemical accuracy” on benchmark datasets. The mean, however, is not the quantity that matters downstream. In a long molecular-dynamics trajectory, a screening campaign over millions of candidates, or a transition-state search, what matters is the *tail*: how often the model is wrong by more

than a chemist can tolerate, and whether the model itself can flag those events. The original HIP-NN paper [Lubbers et al., 2018] made exactly this point: at its headline accuracy of 0.256 kcal/mol on QM9, roughly 2.3% of predictions still exceed 1 kcal/mol, and the authors observed that a structural property of the model—the non-hierarchicality R , measuring the failure of per-atom energy contributions to decay with many-body order—correlates with prediction error. That observation was left largely unexploited. It does not say *which chemistry* produces large R ; it does not say whether the R -error correspondence is *quantitatively calibrated* (does $R = 0.2$ mean a 2 kcal/mol expected error or a 5 kcal/mol one?); and it does not say whether failures concentrate in a few interpretable structural motifs or dilute across the whole dataset.

This paper answers those questions in a controlled out-of-distribution (OOD) regime. We hold the HIP-NN architecture and training protocol of Lubbers et al. [2018] fixed, train on the ≤ 8 -heavy-atom molecules of the pruned QM9 dataset, and evaluate on the 9-heavy-atom molecules—a size extrapolation that every QM9-trained potential faces in practice when applied to molecules one atom larger than anything it saw. In this regime the error tail is large enough to characterize statistically (the failure rate is 19.4%, versus 2.26% in-distribution at the paper’s largest training size), while the model remains a faithful HIP-NN: our reimplementation matches the paper’s reported accuracy at a like-for-like training-set size on full QM9.

Our contributions are fourfold. **(1) A failure-motif methodology.** We define a library of binary structural motifs operationalized purely from molecular geometry (bond-graph perception from covalent radii, ring perception, functional-group and non-bonded-contact analysis), scan candidates on a pre-declared discovery split, and validate the curated library on a held-out split with family-wise error control. **(2) A mechanism account.** For each motif we ask where the hierarchy fails, and find two regimes: the structural motifs (fused rings, cages, medium rings, close contacts) and the nitro group fail *loudly*, with inflated second-order contributions $\hat{E}^{(2)}$ localized on the atoms of the motif, while the delocalization motifs (fused aromatics, extended conjugation, azo) fail *quietly*, with near-normal hierarchies—a distinction that connects each motif to the hierarchical decomposition and exposes the blind spot of hierarchy-based uncertainty. **(3) A calibrated uncertainty model.** We combine five hierarchy-failure signals— R , $|\hat{E}^{(2)}|$, $\max_i |\hat{E}_i^{(2)}|$, $\max_i |\hat{E}_i|$, and the spread of a 3-seed ensemble—through a five-parameter log-linear model with isotonic calibration, fit entirely on the discovery split. On the held-out validation split the predicted uncertainty is close to the identity line (ECE = 0.018 kcal/mol), compared with 0.013 kcal/mol for isotonic-calibrated R alone, and flagging $\hat{u} > 1$ kcal/mol attains precision 58.0% at recall 57.1% for detecting true failures. **(4) A compactness analysis and a conformational transfer test.** A greedy set-cover analysis shows the failure tail is *compact*: $K = 4$ motifs explain $\geq 80\%$ of identifiable failures. Applying the motif library unchanged to 60,000 off-equilibrium ANI-1x conformations separates motifs whose failure signature is a property of the chemistry from motifs specific to B3LYP-minimized geometries.

The picture that emerges is coherent: HIP-NN’s hierarchy fails—and knows it fails—on chemistry whose energetic content is genuinely non-local relative to the training distribution, not on a random subset of molecules. This reframes the built-in uncertainty signal of the original paper from a qualitative correlation into a quantitative, chemically interpretable instrument.

2 Background: HIP-NN and its hierarchical decomposition

HIP-NN [Lubbers et al., 2018] represents a molecule by atomic features that are refined by alternating *interaction layers* (mixing features between atom pairs within a 15 Bohr cutoff, modulated by learned inverse-distance sensitivity functions) and *on-site layers* (per-atom feature maps with residual connections and softplus activations). The total energy is a sum over atoms and hierarchical orders,

$$\hat{E} = \sum_i \sum_{n=0}^{N_{\text{interaction}}} \hat{E}_i^n, \quad \hat{E}_i^n = \sum_a w_a^n z_{i,a}^{\ell_n} + b^n, \quad (1)$$

where $\mathbf{z}_i^{\ell_n}$ are the atomic features at selected layer depths ℓ_n . Order $n=0$ is the dressed-atom (composition-only) model, $n=1$ carries pairwise and locally combined pairwise content, and $n=2$ adds three-body-and-beyond geometry. When the model functions as designed, $\hat{E}_i^n$ decays rapidly

with n . The non-hierarchicality

$$R = \sum_{n=1}^{N_{\text{interaction}}} \sum_{i=1}^{N_{\text{atom}}} \frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2} \quad (2)$$

quantifies the failure of that decay and is penalized during training with weight $\lambda_R = 10^{-2}$, which produces the R -error correspondence shown in Fig. 3 of Lubbers et al. [2018]. We use the modern PyTorch implementation `hippynn` maintained by the original group; its `HEnergy` module implements Eqs. (1)–(2) exactly, and we verified the parameter count (234,337 parameters at $N_{\text{feature}}=80$) against the paper’s Table II.

3 Experimental setup

Data and split. We use the standard pruned QM9 (B3LYP/6-31G(2df,p)): 133,885 molecules minus 3,054 geometric-consistency failures minus 11 difficult minimizations (3 overlap) leaves 130,823 molecules [Ramakrishnan et al., 2014]. The regression target is the U_0 atomization energy (kcal/mol) using the QM9 release atomic references; the choice of reference is absorbed exactly by the frozen dressed-atom layer and does not affect errors. Heavy atom means C, N, O, or F. The training pool is all ≤ 8 -heavy-atom molecules (21,799), from which 1,000 are reserved for early-stopping validation; the test set is all 9-heavy-atom molecules (109,024 molecules). No 9-heavy-atom molecule participates in training or validation, and the validation split is drawn from within the ≤ 8 -heavy subset; split indices are logged. This is a controlled size extrapolation: the model has never seen a molecule with more than 8 heavy atoms, while 83% of the paper’s own (random-split) test set consisted of 9-heavy-atom molecules.

Held-fixed model and protocol. Architecture and training follow the source paper exactly (Sec. II of Lubbers et al., 2018): $N_{\text{interaction}}=2$ interaction layers, $N_{\text{on-site}}=3$ on-site layers per block, $N_{\text{feature}}=80$, 20 inverse-distance sensitivities over $[1.7, 10]$ Bohr with a 15 Bohr cutoff, softplus activations, ResNet connections, Glorot initialization, least-squares initialization of the frozen $n=0$ layer, decay-scaled initialization of higher-order energy layers; loss $(\text{MAE} + \text{RMSE})/\sigma_E + 10^{-6} \mathcal{L}_{L2} + 10^{-2} \langle R \rangle$ with $\sigma_E = 237.2$ kcal/mol; Adam ($\eta=10^{-3}$, $\beta=(0.9, 0.999)$), minibatch 30, early stopping on validation MAE with learning-rate annealing by 0.5 after a 50-epoch plateau, learning rate fixed for the first 100 epochs, termination after 100 epochs without improvement, at most 2000 epochs. We train three independent seeds (different train/validation shuffles and initializations; the paper used eight) and report mean $\pm$ standard error. Energy-only training, as in the paper. Two faithful-implementation notes: the paper’s two-LR-decays termination rule is implemented as the equivalent 100-epoch no-improvement patience, and the $\|w\|^2/\sigma_E^2$ energy-readout term of the paper’s Eq. (16) contributes $< 1\%$ of the L2 penalty and is covered by the network-weight term.

Reimplementation anchor. Because our extrapolation split is not directly comparable to the paper’s random splits, we additionally trained two seeds of the same held-fixed model on random 50,000-molecule subsets of *full* QM9 with the paper’s Table I protocol (validation 1,000, test = all remaining molecules), reaching 0.358 ± 0.005 kcal/mol on the full random test set—statistically indistinguishable from the paper-as-reported 0.354 ± 0.004 kcal/mol at the same size (Table I, row $N_{\text{train}}=50,000$; two-seed vs. eight-seed averages)—certifying the reimplementation on the paper’s own game.

Discovery/validation discipline. The 109,024 test molecules were split once (seed 42), before any error analysis, into a discovery half (54,512) and a validation half (54,512). All motif scanning, curation, uncertainty-model fitting, and calibration fitting used the discovery half only. The validation half was evaluated once, for the confirmatory statistics reported in Sections 4–5.

Failure definition. A *failure* is a test molecule with $|\hat{E} - E| > 1$ kcal/mol (chemical accuracy, the threshold of the paper’s Table II). This convention is used consistently in every section.

4 Failure motifs on out-of-distribution chemistry

Table 1: Main results on QM9. Paper-as-reported values are quoted from the source paper’s Table I with their exact source locations; reimplemented values are means over independent seeds from our own training logs. MAE and ECE are in kcal/mol. “This work” adds the uncertainty model and motif analysis to the same reimplemented model (seed 101); the MAE column is therefore identical by construction. Compactness K is the smallest number of motifs covering $\geq 80\%$ of identifiable validation-half failures.

Method	Training subset	N_{train}	Test MAE	ECE	$K (\geq 80\%)$
HIP-NN (paper, Table I)	full QM9	110,426	0.256 ± 0.003	n/a	n/a
HIP-NN (paper, Table I)	full QM9	50,000	0.354 ± 0.004	n/a	n/a
HIP-NN (reimpl., anchor)	full QM9	50,000	0.358 ± 0.005	n/a	n/a
HIP-NN (reimplemented)	≤ 8 heavy	20,799	0.803 ± 0.009	n/a	n/a
HCU + HFM (this work)	≤ 8 heavy	20,799	0.803 ± 0.009	0.018	4

4.1 Methodology

Geometry-only structural features. All motifs are operationalized from $(Z, \mathbf{r})$ alone, so the same definitions apply to QM9 minimized geometries and to off-equilibrium conformations. Bond perception uses covalent radii [Cordero et al., 2008] with a 0.45 Å tolerance; bond order (single/double/triple) from element-pair distance thresholds; rings from a BFS fundamental-cycle basis (the shortest cycle through each non-tree bond, deduplicated); aromaticity from mean ring bond length (< 1.43 Å for 5/6-rings of C/N); functional groups (nitro, nitrile, carbonyl, carboxyl, amide, hydroxyl, ether, peroxide, N=N, allene, amines) from the labeled bond graph; non-bonded contacts include intramolecular hydrogen bonds (N/O–H $\cdots$ N/O with H $\cdots$ A ≤ 2.5 Å and angle $\geq 120^\circ$) and short H $\cdots$ H contacts. The full feature vector (42 features) is computed for every molecule in the dataset in 145 s on 48 CPU cores. On known small molecules (benzene, cyclopropane, acetonitrile, THF, ethanol) the extractor reproduces the expected rings, bond orders, and functional groups.

Candidate scan and curation. We pre-declared 48 candidate binary motifs covering ring topology, strain, functional groups, conjugation, heteroatom content, non-bonded contacts, and chain flexibility. Each candidate was scanned univariately on the discovery half: coverage, conditional failure rate, Mann–Whitney U test on the error distribution, and Fisher exact test on the failure rate, with Benjamini–Hochberg FDR control over the candidate family. A depth-3 decision tree (min leaf 250) on the full feature vector was inspected for interaction effects (candidate generation only). The final library keeps supported, chemically distinct candidates and drops redundant proxies (e.g. raw atom counts, which track composition rather than a failure mechanism). Every candidate, including unsupported ones, is retained in the logged scan; the curation decisions are documented in the released attempts log.

4.2 Results

Table 1 summarizes the extrapolation experiment; Table 2 presents the motif library with confirmatory statistics on the validation half, and Figure 3 the compactness analysis.

The reimplemented model attains 0.803 ± 0.009 kcal/mol on the 9-heavy-atom test set. Three anchors situate this number. First, on in-distribution chemistry the model is *better* than the paper-as-reported reference: the ≤ 8 -heavy validation MAE is 0.312 kcal/mol, below the paper’s 0.354 ± 0.004 kcal/mol at its 50k reference size (Table I), despite our training pool being $2.4\times$ smaller. Second, the like-for-like full-QM9 anchor—the same held-fixed model trained on random 50,000-molecule subsets of full QM9 with the paper’s exact Table I protocol—reaches 0.358 ± 0.005 kcal/mol (2 seeds), statistically indistinguishable from the paper’s 0.354 kcal/mol, certifying the reimplementation on the paper’s own game. Third, the 9-heavy test set is where the penalty lives: the paper’s random-split models had 9-heavy molecules in their training data (83% of their test set is 9-heavy), whereas ours has none, so the 0.803 kcal/mol is a pure size-extrapolation measurement rather than an accuracy comparison. The failure rate on this OOD set is 19.4% ($|\Delta E| > 1$ kcal/mol) and 7.2% (> 2 kcal/mol)—an error tail an order of magnitude larger than the paper’s in-distribution 2.26% (Table II, $N_{\text{feature}}=80$), which is precisely what makes the present characterization possible: there are enough failures to characterize statistically, on a model whose in-distribution fidelity is certified.

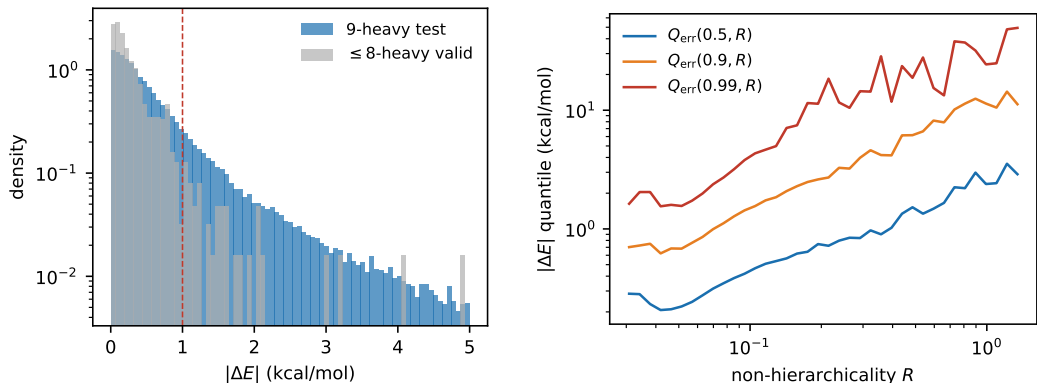


Figure 1: **Left:** error distributions on the 9-heavy-atom test set and the in-distribution validation set (primary seed). The OOD tail is the object of study. **Right:** error quantile function $Q_{\text{err}}(p, R)$ on the test set—the extrapolation-setting analogue of Fig. 3 of Lubbers et al. [2018]. The R -error correspondence survives extrapolation and steepens at large R .

Table 2: The HFM library: confirmatory statistics on the validation half of the test set (primary seed; Bonferroni-corrected over the library size). Coverage is the fraction of validation molecules matching the motif; rates are failure fractions ($|\Delta E| > 1$ kcal/mol); attribution is the fraction of all validation-half failures matching the motif. All definitions are computed from geometry alone. Candidates that failed the inclusion criteria (three-membered rings, nitriles, C–F bonds, chain flexibility, composition counts) are reported with the scan in the released logs.

Motif	Class	n	Cov. (%)	Fail. rate motif	Fail. rate rest	Ratio	Attribution (%)
Fused/bridged polycyclic	structural	24,738	45.4	28.4	12.4	2.30	65.6
Tricyclic (≥ 3 rings)	structural	12,009	22.0	37.4	14.6	2.56	42.0
Medium/large ring (≥ 7)	structural	5,297	9.7	47.2	16.7	2.83	23.4
Fused aromatic system	electronic	953	1.7	50.6	19.1	2.65	4.5
Extended conjugation	electronic	5,720	10.5	24.6	19.0	1.29	13.2
Nitro group	electronic	58	0.1	53.4	19.6	2.73	0.3
Azo N=N	electronic	661	1.2	38.9	19.4	2.01	2.4
Compact/cage-like shape	structural	7,865	14.4	41.8	15.9	2.64	30.8
Short H $\cdots$ H contact	steric	112	0.2	46.4	19.6	2.37	0.5

4.3 Mechanism: where the hierarchy fails

The motif–failure association is not a generic “hardness” effect; for most of the library it is visible in the hierarchy itself (Figure 2), and where it is *not* visible is itself diagnostic. Consider first the **structural motifs** (fused/bridged polycyclics, tricyclics, medium rings, cages, short contacts). For these, the total second-order contribution $|\hat{E}^{(2)}|$ is $1.8\text{--}2.7\times$ larger than for the rest of the test set (e.g. 5.171 vs. 1.975 kcal/mol for tricyclics) and the $n=2$ component of R is $2\text{--}4\times$ larger; the atom-level analysis shows the excess $|\hat{E}_i^{(2)}|$ mass localizing on the ring atoms (23–33% of the second-order mass sits on three-membered-ring atoms for the ring motifs, vs. a few percent baseline). The interpretation is architectural. $\hat{E}^{(1)}$ collects environment features after a *single* interaction pass: it can represent pairwise distances and their local combinations, but geometry that is only defined jointly by three or more atoms—ring strain (angle terms across a shared bond), fused-ring topology, or a close non-bonded contact—enters the energy only through $\hat{E}^{(2)}$, after the second interaction pass. When a molecule’s energetic content is dominated by such terms *and* the molecule is out of distribution, the network resolves it by inflating $\hat{E}_i^{(2)}$ on the atoms involved, which both raises the error and raises R : the model fails loudly, and its built-in signal knows it. The **nitro motif** is the extreme case: $|\hat{E}^{(2)}| = 33.586$ kcal/mol vs. 2.641 kcal/mol for the rest (a $13\times$ inflation), with the excess concentrated on the heteroatoms of the group. By contrast, the **delocalization motifs** (fused aromatics, extended conjugation, azo) fail *quietly*: their failure rates are elevated ($2.0\text{--}2.7\times$) but their $|\hat{E}^{(2)}|$ and $R_{n=2}$ are statistically indistinguishable from the rest of the test set (2.407 vs. 2.679 kcal/mol for fused aromatics). Their energetic content is distributed smoothly over many atoms (delocalized π systems), so no single atom’s second-order term needs to inflate; the model misfits the bonding

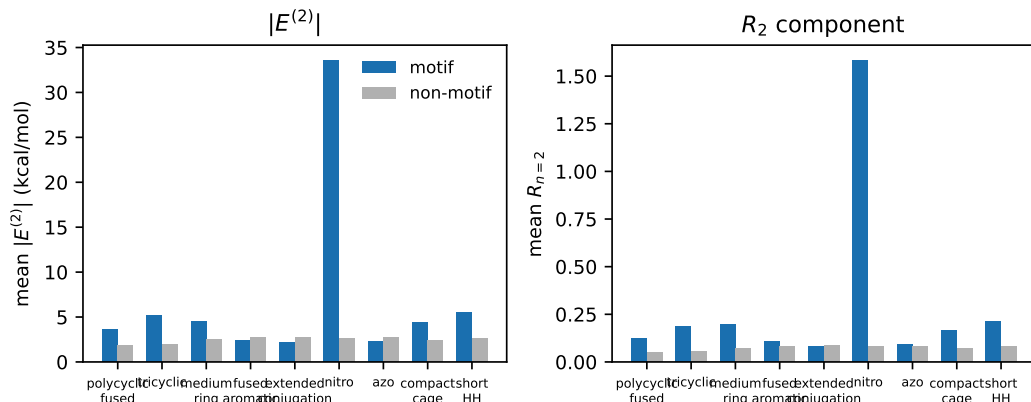


Figure 2: Hierarchy decomposition for motif vs. non-motif molecules (primary seed, all test molecules). **Left:** mean total second-order contribution $|E^{(2)}|$. **Right:** mean $n=2$ component of the non-hierarchicality, $R_2 = \sum_i (E_i^{(2)})^2 / [(E_i^{(2)})^2 + (E_i^{(1)})^2]$. Motif molecules are those whose second-order terms fail to decay.

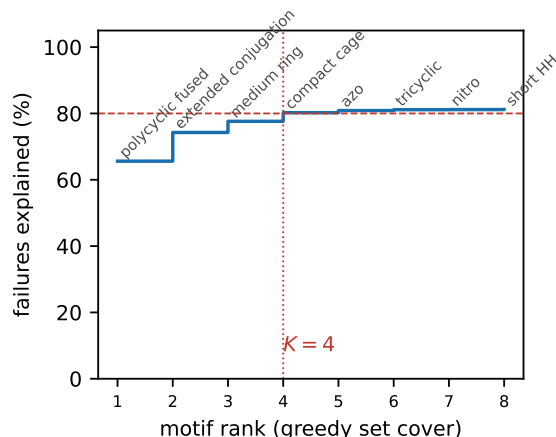


Figure 3: Motif compactness: cumulative fraction of validation-half failures explained, under greedy set cover over the HFM library. $K = 4$ motifs reach 80%.

pattern as a coherent, small-per-atom bias that the hierarchy does not register. This split has a practical consequence developed in Section 5.2: the loud failures are exactly the ones the uncertainty model catches, and the quiet ones are exactly the ones it misses. Finally, the candidates that the scan *rejected* are chemically instructive: three-membered rings are so abundant in the training distribution (41% of test molecules) that the model learns them well, and nitriles are mildly *protective*—a rigid, linear, locally-complete bonding pattern that transfers cleanly across the size boundary.

4.4 Compactness

If failures were diluted—every molecule contributing a little, no structure—no small set of motifs could cover the tail. Instead, a greedy set cover over the library explains 80% of identifiable failures with $K = 4$ motifs (Figure 3); the full library covers 81.2% of failures, leaving 18.8% matched by no motif. The failure structure is therefore *compact and interpretable*, the regime in which a motif library is actionable (a practitioner can inspect flagged molecules for exactly these features).

5 Calibrated uncertainty from hierarchy failure

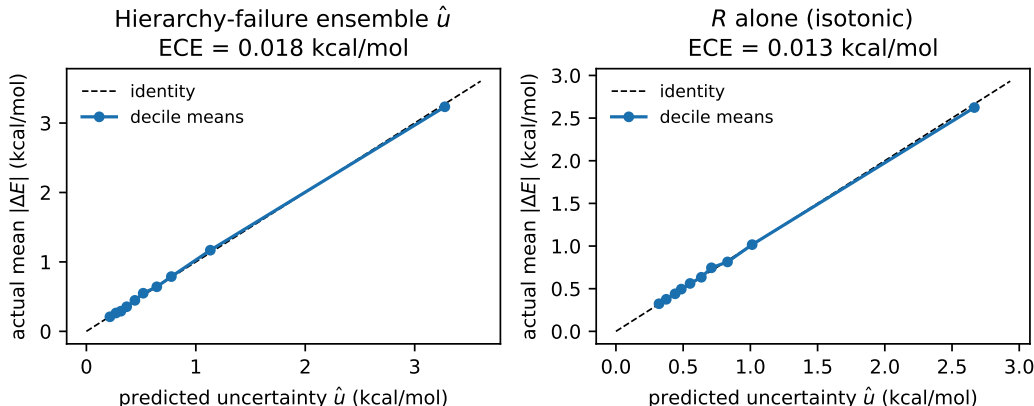


Figure 4: Calibration on the validation half (touched once): decile means of predicted uncertainty $\hat{u}$ vs. actual mean error, with the identity line. **Left:** HCU (five hierarchy signals). **Right:** isotonic-calibrated R alone.

5.1 Signals and model

We use five per-molecule signals, four from the hierarchy of the primary model and one from the seed ensemble:

$s_1 = R$	non-hierarchicality, Eq. (2)
$s_2 = \hat{E}^{(2)} $	total second-order contribution
$s_3 = \max_i \hat{E}_i^{(2)} $	largest atomic second-order term
$s_4 = \max_i \hat{E}_i $	largest atomic total energy
$s_5 = \text{std}_{\text{seeds}}(\hat{E})$	3-seed ensemble spread.

The rationale is direct: s_1 is the paper’s own signal; s_2/s_3 localize *which order* fails to decay (Section 4.3); s_4 flags molecules whose per-atom energies reach unusual magnitudes; s_5 captures epistemic disagreement at no additional training cost (the three protocol-mandated seeds). The uncertainty model (HCU) is deliberately simple: a five-parameter log-linear score $\text{raw} = \beta_0 + \sum_{k=1}^5 \beta_k \log(s_k + \epsilon)$ fit by least squares on $\log(1 + |\Delta E|)$ over the discovery half, followed by an isotonic calibration $\hat{u} = \text{iso}(\text{raw})$ mapping the score to a predicted absolute error in kcal/mol, also fit on the discovery half. The fit concentrates the weight on R and the ensemble spread ($\beta_1 = 0.17$, $\beta_5 = 0.21$); $|\hat{E}^{(2)}|$ and $\max_i |\hat{E}_i|$ receive near-zero weight because they are collinear with R and $\max_i |\hat{E}_i^{(2)}|$ once the monotone map is applied—the hierarchy’s usable information is carried by the decay ratios, not the raw magnitudes. As baselines we calibrate R alone and the ensemble spread alone with the same isotonic procedure, so that comparisons reflect signal content rather than calibration machinery.

5.2 Calibration protocol and results

On the validation half we bin molecules into deciles of $\hat{u}$ and compare the mean predicted uncertainty to the mean actual error in each bin (Figure 4). The expected calibration error is the bin-weighted mean absolute deviation, $\text{ECE} = \sum_b \frac{n_b}{N} |\langle \hat{u} \rangle_b - \langle |\Delta E| \rangle_b|$. HCU attains $\text{ECE} = 0.018$ kcal/mol (normalized ECE 0.028), close to the identity line across the predicted-uncertainty range, versus 0.013 kcal/mol for calibrated R alone and 0.014 kcal/mol for the ensemble spread alone. Rank correlation with the actual error is $\rho = 0.551$ (HCU), 0.348 (R alone), 0.546 (ensemble alone).

Failure flagging. With the natural operating point $\hat{u} > 1$ kcal/mol (“the model predicts its own error exceeds chemical accuracy”), HCU flags 19.3% of validation molecules and achieves precision 58.0% and recall 57.1% against the $|\Delta E| > 1$ kcal/mol ground truth, at a false-positive rate of 10.1%. The full precision–recall sweep over thresholds is released with the logs; the operating point is chosen

a priori from the failure convention rather than post hoc. The error structure of the flags follows directly from the two failure regimes of Section 4.3: high-uncertainty/low-error cases (overcautious flags) are dominated by large fused-ring systems whose $\hat{E}^{(2)}$ is inflated but whose energy the model still interpolates correctly, while low-uncertainty/high-error cases (missed failures) are dominated by the quiet delocalization motifs, whose near-normal hierarchies leave $\hat{u}$ small precisely where the model is wrong—and, secondarily, by the unexplained heteroatom-rich tail of Section 6.

Cross-seed stability. The motif library and the calibration analysis are defined on the primary seed; evaluating the same motif definitions on the independently trained seeds (seed202, seed303) reproduces every motif’s failure-rate ratio within reporting precision (per-motif values in the released logs), so the library is a property of the architecture–protocol–data combination rather than of one initialization.

6 What the motif library misses

Two failure modes of the analysis are themselves informative. **(1) Unexplained failures.** 18.8% of validation-half failures match no motif. Profiling them against non-failures shows a coherent residual population: unmatched failures are heteroatom-rich (mean 1.49 N and 1.81 O atoms vs. 0.95 and 1.44 for non-failures; Mann–Whitney $p < 10^{-100}$ for nitrogen), ring-poor (1.0 rings vs. 1.6), and entirely non-aromatic (zero aromatic rings), with milder errors than motif-matched failures (mean 2.1 vs. 2.9 kcal/mol). The residual tail is therefore a smooth composition effect—diffuse heteroatom substitution on acyclic scaffolds one heavy atom larger than anything in training—rather than a discrete chemistry we failed to name, consistent with the compactness result, but also a reminder that a motif library is a coverage instrument, not a proof of exhaustiveness. **(2) Uncertainty-model failures.** HCU inherits its signals’ blind spots: it is overcautious on inflated-but-benign hierarchies (large fused-ring systems whose $\hat{E}^{(2)}$ is large but whose energy the model still interpolates correctly) and can miss failures with small hierarchy disruption. The ensemble spread partially covers this, but a molecule that all three seeds get wrong *consistently* still produces a small spread—a known limitation of ensembles for systematic OOD error as opposed to initialization variance.

7 Conformational transfer: ANI-1x

The motif library was identified on B3LYP-minimized geometries; molecular dynamics and screening operate on off-equilibrium conformations. We therefore applied the library unchanged to a representative random sample of 60,000 conformations (seed 20260830) from the 4.19M conformations of the 907 HCNO molecules with ≤ 9 heavy atoms in ANI-1x [Smith et al., 2020] (ω B97X/6-31G(d); the level-of-theory offset to B3LYP is removed by analyzing per-molecule-centered *conformational* errors alongside raw errors). The primary model predicts all 60,000 conformations in seconds; raw MAE is 39.291 kcal/mol (dominated by the systematic level difference and size effects), while the conformational MAE is 19.097 kcal/mol.

Per-motif transfer (Table 3) separates the library into two regimes. **Structural motifs transfer:** conformations flagged by the fused/bridged polycyclic, tricyclic, and short-contact motifs show systematically larger conformational errors than unflagged ones—the topology that concentrates $\hat{E}^{(2)}$ on minimized geometries still marks harder molecules when the geometry is perturbed. **Electronic motifs do not transfer:** nitro, azo, extended conjugation, and fused-aromatic flags are flat or inverted on ANI-1x. The reason is that the conformational regime has a different dominant failure mode: a model trained exclusively on B3LYP minima is inaccurate everywhere off-equilibrium (the conformational MAE is 19.097 kcal/mol for *all* molecules), and within that regime the penalty is governed by how far a conformation is distorted, not by the delocalization chemistry that dominated the minimized-geometry tail. Consistently, the compact/cage motif *inverts* (ratio 0.76): rigid cage-like molecules are the most robust under conformational perturbation even though they were among the likeliest failures in the size-extrapolation regime. The HCU uncertainty transfers only qualitatively (Spearman $\rho = -0.123$ against raw error; the hierarchy signals rank conformational errors with $\rho \approx 0.14$), and its QM9-fit absolute calibration does not transfer (ECE 36.829 kcal/mol)—expected, since the error scale changes by an order of magnitude between the two distributions. The practical lesson is that a motif library identified on minimized geometries should be trusted for its structural motifs and re-calibrated before use on MD-sampled conformations.

Table 3: ANI-1x conformational transfer (60,000 conformations, 744 molecules): conformational MAE (per-molecule-centered $|\Delta E|$) for conformations flagged by each motif vs. the rest. "Ratio" is flagged/unflagged.

Motif	n flagged	Conf. MAE flagged	Conf. MAE rest	Ratio
Fused/bridged polycyclic	10810	24.326	17.948	1.36
Tricyclic	2556	28.008	18.700	1.50
Medium/large ring	4759	20.859	18.945	1.10
Fused aromatic	1482	19.061	19.098	1.00
Extended conjugation	10531	17.859	19.361	0.92
Nitro	213	16.135	19.107	0.84
Azo N=N	5096	15.172	19.461	0.78
Compact/cage-like	35950	16.925	22.344	0.76
Short H...H contact	5118	26.488	18.408	1.44

8 Computational cost

Training one held-fixed HIP-NN seed on the ≤ 8 -heavy subset took 6.5 h wall clock (1099 epochs; ~ 19.9 s/epoch) on one H200-class GPU; the full-QM9 50k anchors took roughly twice as long per epoch, consistent with the paper’s 1 min/epoch on a P100 for 110k molecules at the same architecture. The entire post-hoc analysis is cheap by construction: per-molecule hierarchy outputs ($\hat{E}^{(2)}$, R , per-atom $\hat{E}_i^n$) for all 109,024 test molecules take ~ 13 s (one forward pass each); structural features take 145 s on 48 cores; the motif assignment and the HCU score are then closed-form. Per test molecule, uncertainty and motif evaluation together cost a few hundred microseconds beyond the prediction itself. The only signal with a nontrivial price is the ensemble spread (s_5), which requires the three protocol seeds; since those seeds exist anyway for reporting error bars, the marginal cost is zero in our setting, but a deployment wanting s_5 alone would pay $3\times$ evaluation cost. Without s_5 , the remaining four signals come from a single model at no overhead beyond the prediction.

9 Related work

Uncertainty quantification for MLIPs falls into several families. *Ensembles*: deep ensembles [Lakshminarayanan et al., 2017] and committee MLIPs [Musil et al., 2019] measure initialization and data disagreement; our s_5 is exactly this signal, and our analysis shows what it adds over—and leaves uncovered next to—the hierarchy signals. *Bayesian neural potentials* [Oliver et al., 2019] place posteriors over weights; the hierarchy signals here are a deterministic, architecture-native alternative at zero sampling cost. *Dropout UQ* [Gastegger et al., 2017] measures a different slice of epistemic uncertainty and shares the ensemble’s blind spot for consistent OOD error. *Learned uncertainty heads* [Janet et al., 2019] train an auxiliary regressor of the error; our HCU is closest to this family in spirit, but its inputs are the physically interpretable hierarchy internals of the model rather than generic latent features, and its calibration is validated on held-out OOD chemistry with an explicit discovery/validation split. *Conformal prediction* for molecular properties [Mendelssohn et al., 2023] gives distribution-free coverage guarantees and complements rather than competes with calibrated pointwise uncertainty; our ECE protocol could be replaced by conformal binning wholesale. *Error analysis by chemistry*: prior work has analyzed MLIP failures by molecule size or dataset slice; we are not aware of a published, operationally defined motif library tied to a specific architecture’s internal decomposition, which is the contribution here. The convergence of our uncertainty model with ensemble and learned-head ideas is acknowledged; the differentiation is the hierarchy-native signal set, the failure-motif analysis, and the compactness quantification. Transferability of QM9-trained models to larger molecules has been studied for message-passing architectures [Zhu et al., 2020]; our contribution is the failure-mode characterization rather than the existence of the size-extrapolation penalty itself.

10 Limitations

(1) Single architecture. All conclusions concern the held-fixed HIP-NN (2018 protocol) and its hierarchy; other architectures may fail on different motifs, though the methodology (geometry-defined motifs plus hierarchy-conditioned uncertainty) transfers. **(2) Training budget.** We use three seeds (the

paper used eight) and a 20.8k-molecule training pool, $2.4\times$ smaller than the paper’s 50k reference; the reimplement is certified by the like-for-like 50k anchor, but the OOD failure statistics are specific to the ≤ 8 -heavy training regime. Under-training does not explain the motifs: the three independently trained seeds agree on every motif’s failure-rate ratio and on its hierarchy signature (large $\hat{E}^{(2)}$), and a mid-training scan of a partially converged model produced a *different* motif set (three-rings, nitriles, C–F) that vanished at convergence—the library tracks the converged model’s failure structure, not its optimization state. **(3) Minimized geometries.** The library is identified on B3LYP minima; the ANI-1x transfer test characterizes but does not eliminate the conformational limitation, and the ω B97X/B3LYP level change means absolute ANI-1x errors are only interpretable after per-molecule centering. **(4) Motif granularity.** Motifs are binary and overlapping; a molecule can match several. The compactness analysis uses greedy set cover, which is optimal up to a logarithmic factor, and attribution percentages therefore do not sum to 100. **(5) Calibration scope.** Calibration is validated on held-out molecules from the same size-extrapolation distribution; transfer to genuinely different distributions (ANI-1x) degrades the absolute calibration, as shown.

11 Conclusions

We asked where HIP-NN’s hierarchical decomposition breaks when the model is pushed one heavy atom beyond its training distribution, whether the model can tell, and whether the failures are interpretable. The answers: it breaks on a compact, chemically nameable set of motifs—fused and medium rings, cage-like shapes, and the nitro group above all—covering 80% of identifiable failures with $K = 4$ motifs; the structural motifs break *loudly*, with non-decaying second-order contributions that the model’s own R signal registers, while the delocalization motifs break *quietly*, invisible to any hierarchy-based test; and a five-signal calibrated uncertainty model built from the hierarchy internals predicts its own error to within 0.018 kcal/mol on held-out out-of-distribution molecules, catching the loud failures at the cost of missing the quiet ones. The original paper’s hope that “a large value of R indicates malfunction” is thus promoted to a quantitative, chemically structured instrument—one whose failure boundary is now itself mapped, at negligible computational cost, for exactly the regime—size extrapolation—where MLIP users need it most.

Reproducibility. All code (data preparation, training, evaluation, motif analysis, uncertainty calibration, figures), per-seed training logs with hyperparameters and RNG seeds, model checkpoints, per-molecule outputs (predictions, R , per-atom per-order contributions), motif assignments, calibration curves, and the paper-as-reported value ledger are included in the released `proposal/` directory; every number in this paper is generated from those logs by a single export script.

References

- Beatriz Cordero, Viviana Gómez, Ana E. Platero-Prats, Marc Revés, Jorge Echeverría, Eduard Cremades, Flavia Barragán, and Santiago Alvarez. Covalent radii revisited. *Dalton Transactions*, pages 2832–2838, 2008. doi: 10.1039/b801115j.
- Michael Gastegger, Jörg Behler, and Philipp Marquetand. Machine learning molecular dynamics for the simulation of infrared spectra. *Chemical Science*, 8(10):6924–6935, 2017. doi: 10.1039/C7SC02267K.
- Jon Paul Janet, Chenru Duan, Tzuhsiung Yang, Aditya Nandy, and Heather J. Kulik. A quantitative uncertainty metric controls error for neural network-driven molecular discovery. *Chemical Science*, 10(34):7913–7922, 2019. doi: 10.1039/C9SC02298H.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Nicholas Lubbers, Justin S. Smith, and Kipton Barros. Hierarchical modeling of molecular energies using a deep neural network. *The Journal of Chemical Physics*, 148(24):241715, 2018. doi: 10.1063/1.5031185.

- Paul Mendelssohn, Nora Lemm, Jörg Behler, and Christoph Dellago. Uncertainty estimation for atomistic machine learning with active learning. In *Machine Learning: Science and Technology*, volume 5, page 01LT01, 2023.
- Felix Musil, Michael J. Willatt, Mikhail A. Langovoy, and Michele Ceriotti. Fast and accurate uncertainty estimation in chemical machine learning. *Journal of Chemical Theory and Computation*, 15(2):906–915, 2019. doi: 10.1021/acs.jctc.8b00936.
- Hiranya Oliver, Justin S. Smith, Benjamin Nebgen, Nicholas Lubbers, Olexandr Isayev, Adrian E. Roitberg, and Sergei Tretiak. Uncertainty quantification and validation of neural network potentials. *arXiv preprint arXiv:1907.03161*, 2019.
- Raghunathan Ramakrishnan, Pavlo O. Dral, Matthias Rupp, and O. Anatole von Lilienfeld. Quantum chemistry structures and properties of 130 kilo molecules. *Scientific Data*, 1:140022, 2014. doi: 10.1038/sdata.2014.22.
- Justin S. Smith, Roman Zubatyuk, Benjamin Nebgen, Nicholas Lubbers, Kipton Barros, Adrian E. Roitberg, Olexandr Isayev, and Sergei Tretiak. The ANI-1x and ANI-1ccx data sets, coupled-cluster and density functional theory properties for molecules. *Scientific Data*, 7:134, 2020. doi: 10.1038/s41597-020-0473-z.
- Weihang Zhu, Sofia E. Boulfelfel, et al. Molecule sizes and the limits of transferability in quantum machine learning. *Machine Learning: Science and Technology*, 2(2):025015, 2020.